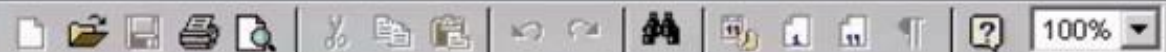


Question No : 41 of 52

State the sufficient condition of convergence of the iterative solution to the exact solution

Answer ([Please click here to Add Answer](#))



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Answer: A sufficient condition for convergence of iterative solution to exact solution is $|a_1| \Rightarrow (\text{greater or equal}) |b_1| + |c_1|$ $|b_2| \Rightarrow (\text{greater or equal}) |a_2| + |c_2|$ $|c_3| \Rightarrow (\text{greater or equal}) |a_3| + |b_3|$ For the system $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$ Similarly for the system with more variables we can also construct the same condition

Question No : 42 of 52

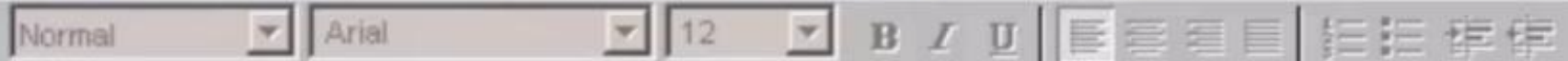
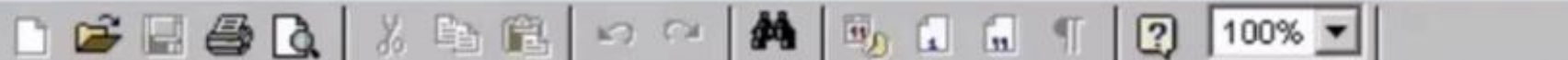
Evaluate the integral

$$\int_3^5 (\log x + 1) dx$$

Using Trapezoidal rule

Take $h=1$

Answer ([Please click here to Add Answer](#))



X	3	4	5
Y=	0.602	0.698	0.778
logx+1	0	9	1

Using trapezoidal rule, taking $h=1$

$$I = \frac{h}{2} (y_0 + y_3 + 2y_1)$$

$$I = \frac{1}{2} (0.6020 + 0.7781 + 2(0.6989))$$

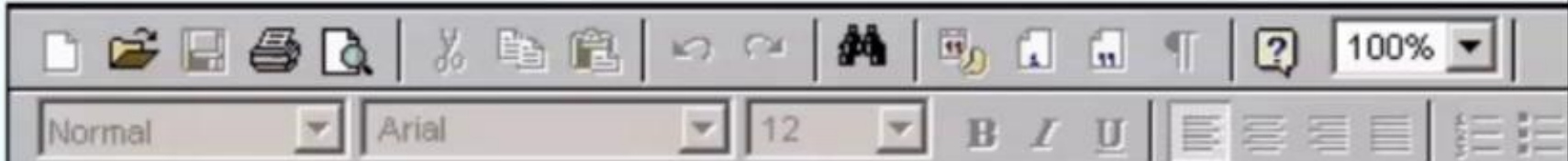
$$I = \frac{1}{2} (1.3801 + 1.3978)$$

$$I = \frac{2.7779}{2} = 1.38895$$

Question No : 43 of 52

Write a formula for finding the value of k_1 in Fourth-order R-K method.

Answer ([Please click here to Add Answer](#))

A rich text editor toolbar with various icons for document manipulation. It includes icons for new, open, save, print, find, undo, redo, bold, italic, underline, bulleted list, numbered list, and a zoom dropdown set to 100%. Below the icons are dropdown menus for text style (Normal), font family (Arial), and font size (12), followed by buttons for bold (B), italic (I), and underline (U), and a section for alignment and indentation.

$$k_1 = hf(t_n, y_n)$$

Question No : 44 of 52

Write Adam-Moulton's Predictor formula for finding the solution of a differential equation.

Answer ([Please click here to Add Answer](#))



$$y_{n+1} = y_n + \frac{h}{24} [55y'_n - 59y'_{n-1} + 37y'_{n-2} - 9y'_{n-3}] + \frac{251}{720} h \nabla^4 y'$$

This is known as Adam's predictor formula.

If $f(2) = -2.6146$ and $f(3) = 4.7610$, then find the first approximation using the Regula-Falsi method

Answer ([Please click here to Add Answer](#))



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$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

$$x_3 = 3 - \frac{3 - 2}{4.7610 - (-2.6146)} (4.7610)$$

$$x_3 = 3 - \frac{4.7610}{7.3756}$$

$$x_3 = 3 - \frac{4.7610}{7.3756} = 3 - 0.6455 = 2.3545$$

$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

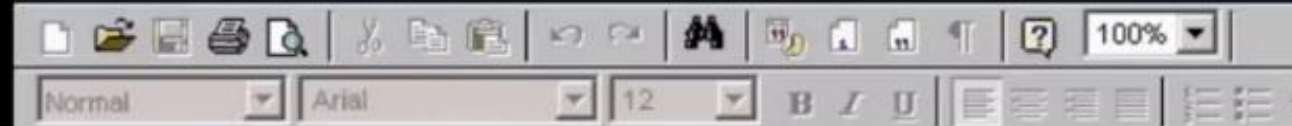
$$x_3 = 3 - \frac{3 - 2}{4.7610 - (-2.6146)} (4.7610)$$

$$x_3 = 3 - \frac{4.7610}{7.3756}$$

$$x_3 = 3 - \frac{4.7610}{7.3756} = 3 - 0.6455 = 2.3545$$

If, in solving a given differential equation, $y_0 = 1, y_1' = 1.1, y_2' = 1.2, y_3' = 1.3, h = 1$
Then find y_4 by Milne's Predictor formula.

Answer ([Please click here to Add Answer](#))



$$y_4 = y_0 + \frac{4h}{3}(2y_1' - y_2' + 2y_3')$$

$$P: y_{n+1} = y_{n-3} + \frac{4h}{3}(2y_{n-2}' - y_{n-1}' + 2y_n')$$

$$P: y_4 = y_0 + \frac{4h}{3}(2y_1' - y_2' + 2y_3')$$

$$P: y_4 = 1 + \frac{4(1)}{3}(2(1.1) - 1.2 + 2(1.3))$$

$$P: y_4 = 1 + \frac{4}{3}(2.2 - 1.2 + 2.6)$$

$$P: y_4 = 1 + \frac{4}{3}(3.6)$$

$$P: y_4 = 1 + \frac{14.4}{3}$$

$$P: y_4 = \frac{3 + 14.4}{3} = \frac{17.4}{3} = 5.8$$

Find $F(h)$, using Richardson's extrapolation limit, while finding $y'(0.01)$ to the function $y = 1/x$ with $h=0.005$.

Answer ([Please click here to Add Answer](#))

Richmond's extrapolation limit formula:

$$F(h) = \frac{y(x+h) - y(x-h)}{2h}$$

$$F(h) = \frac{\frac{1}{0.01+0.005} - \frac{1}{0.01-0.005}}{2(0.005)}$$

$$F(h) = \frac{\frac{1}{0.015} - \frac{1}{0.005}}{0.01}$$

$$F(h) = \frac{66.66 - 200}{0.01}$$

$$F(h) = \frac{-133.34}{0.01} = -13334$$

Evaluate the integral

$$\int_0^3 (x^2 + x) dx$$

Using Simpson's 3/8 rule

Answer ([Please click here to Add Answer](#))

X	0	1	2	3
Y=(x ² +x)	0	2	6	12


 100%



$$\int_0^3 (x^2 + x) dx = \frac{3}{8} h [y_0 + y_3 + 3(y_1 + y_2)]$$

$$\int_0^3 (x^2 + x) dx = \frac{3(1)}{8} [0 + 12 + 3(2 + 6)]$$

$$\int_0^3 (x^2 + x) dx = \frac{3}{8} [12 + 24] = \frac{3}{8} (36) = \frac{108}{8} = 13.5$$

Question No : 50 of 52

Construct a backward difference table from the following values of x and y.

x	-1	0	1	2	3
y=f(x)	10	2	10	62	80

Answer ([Please click here to Add Answer](#))

Rich text editor toolbar with icons for file operations, text formatting, and a dropdown menu showing 'Normal' and 'Arial'.

x	Y=f(x)	∇_y	∇^2_y	∇^3_y	∇^4_y
-1	10				
0	2	-8			
1	10	8	16		
2	62	52	44	28	
3	80	18	-34	-78	-106

Question No : 51 of 52

From the following table of values, construct forward difference table.

x	1.00	1.05	1.10	1.15	1.20
y	1.0000	1.0247	1.0488	1.0724	1.0954

Answer ([Please click here to Add Answer](#))

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x	$Y=f(x)$	Δ_y	Δ^2_y	Δ^3_y	Δ^4_y
1.00	1.000				
1.05	1.0247	0.0247			
1.10	1.0488	0.0241	0.0006		
1.15	1.0724	0.0236	-0.0005	-0.0011	
1.20	1.0954	0.023	-0.0006	-0.0001	0.0000

Question No : 52 of 52

Use Runge-Kutta Method of order four to find the values of k_1, k_2, k_3 and k_4 for the initial value problem

$$y' = \frac{1}{2}(2x^3 + y), y(1) = 2 \text{ taking } h = 0.1$$

Answer ([Please click here to Add](#))

$$\text{here } f(x, y) = \frac{1}{2}(2x^3 + y), \quad h = 0.1, \quad x_1 = 1, \quad y_1 = 2$$

$$K_1 = hf(x_1, y_1) = 0.1 \left(\frac{1}{2}(2(1)^3 + 2) \right)$$

$$K_1 = hf(x_1, y_1) = 0.1 \left(\frac{6}{2} \right) = \frac{0.6}{2} = 0.3$$

$$K_2 = hf \left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2} \right) = 0.1 \left(1 + \frac{0.1}{2}, 2 + \frac{0.3}{2} \right)$$

$$K_2 = hf \left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2} \right) = 0.1(1.05, 2 + 0.15)$$

$$K_2 = hf \left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2} \right) = 0.1(1.05, 2.15) = 0.1(1.05 + 2.15) = 0.32$$

$$K_3 = hf \left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2} \right) = 0.1 \left(1 + \frac{0.1}{2}, 2 + \frac{0.32}{2} \right)$$

$$K_3 = hf \left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2} \right) = 0.1(1.05, 2 + 0.16)$$

$$K_3 = hf \left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2} \right) = 0.1(1.05, 2.16) = 0.1(1.05 + 2.16) = 0.321$$

$$K_4 = hf(x_1 + h, y_1 + k_3) = 0.1(1 + 0.1, 2 + 0.321)$$

$$K_4 = hf(x_1 + h, y_1 + k_3) = 0.1(1.1, 2.321) = 0.1(0.1 + 2.21575) = 0.3421$$

Now,

$$y_2 = y_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$y_2 = 2 + \frac{1}{6}(0.3 + 2(0.32) + 2(0.321) + 0.3421)$$

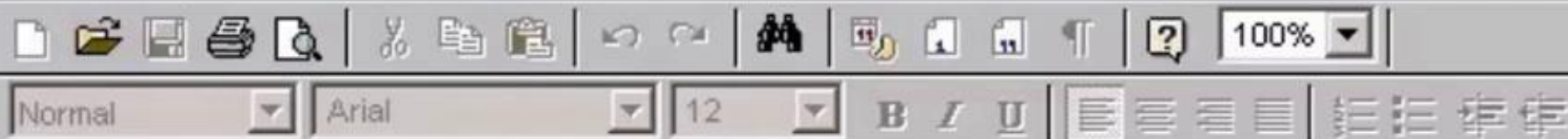
$$y_2 = 2 + \frac{1}{6}(0.3 + 0.64 + 0.642 + 0.3421) = 2 + \frac{1.9241}{6} = 2 + 0.320683 = 2.320683$$

Obtain numerically the solution of

$$y' = x + y^2, \quad y(0) = 2$$

Using Euler's method to find y at $x=1, h=1$

Answer ([Please click here to Add Answer](#))



$$h = 1, x = 1 \text{ and } y(0) = 2$$

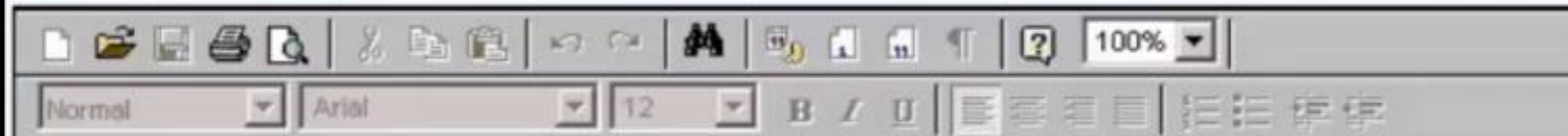
$$y = y_0 + hf(x_0, y_0)$$

$$y = 2 + 1(1 + 2)$$

$$y = 2 + 3 = 5$$

If $f(0) = 3$ and $f(1) = 9$, then find the next approximate value of the function using secant method

Answer ([Please click here to Add Answer](#))



$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = \frac{(0)9 - (1)3}{9 - 3}$$

$$x_2 = \frac{0 - 3}{9 - 3} = \frac{-3}{6} = -0.5$$

Write a backward difference formula of $D^2 f(x)$.

Answer ([Please click here to Add Answer](#))



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$$y_n'' = D^2 y_n = \frac{1}{h^2} \left(\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n + \dots \right)$$

Evaluate the integral

$$\int_3^5 (\log x + 2) dx$$

Using Simpson's 3/8 rule

Take $h=1$

Answer ([Please click here to Add Answer](#))

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X	3	4	5
Y=(log x + x)	3.477	4.602	5.6989
	1	0	

$$\int_3^5 (\log x + x) dx = \frac{3}{8} h (y_0 + y_2 + 3y_1)$$

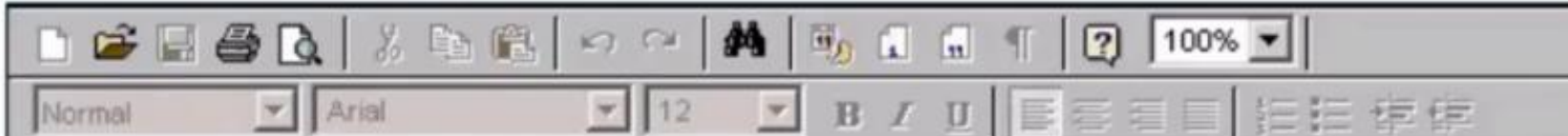
$$\int_3^5 (\log x + x) dx = \frac{3}{8} (1) (3.4771 + 5.6989 + 3(4.6020))$$

$$\int_3^5 (\log x + x) dx = \frac{3}{8} (9.176 + 13.806)$$

$$\int_3^5 (\log x + x) dx = \frac{3}{8} (22.982) = \frac{68.946}{8} = 8.61825$$

Write the two steps of solving the linear equations using Gaussian Elimination method

Answer ([Please click here to Add Answer](#))



In this method, the solution to the system of equations is obtained in two stages.

- i) the given system of equations is reduced to an equivalent upper triangular form using elementary transformations
- ii) The upper triangular system is solved using back substitution procedure

Use Runge-Kutta Method of order four to find the values of k_1 and k_2 for the initial value problem

$$y' = yx^2, y(0) = 1, h = 0.1$$

Answer ([Please click here to Add Answer](#))

$$f(x, y) = yx^2, y(0) = 1 \text{ and } h = 0.1, x_0 = 0$$

$$k_1 = hf(x_0, y_0) = 0.1(1(0^2)) = 0$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= 0.1\left(0 + \frac{0.1}{2}, 1 + \frac{0}{2}\right)$$

$$= 0.1(0.05, 1) = 0.1(0.05 + 1) = 0.105$$

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2)$$

$$y_1 = 1 + \frac{1}{6}(0 + 2(0.105)) = 1 + \frac{0.21}{6} = 1.035$$

Evaluate the integral

$$\int_0^3 (x^2 + x) dx$$

Using Trapezoidal rule

Take $h=1$ Answer ([Please click here to Add Answer](#))

X	0	1	2	3
Y=(x ² + x)	0	2	6	12

$$\int_0^3 (x^2 + x) dx = \frac{h}{2} (y_0 + y_3 + 2(y_1 + y_2))$$

$$\int_0^3 (x^2 + x) dx = \frac{1}{2} (0 + 12 + 2(2 + 6))$$

$$\int_0^3 (x^2 + x) dx = \frac{1}{2} (28)$$

$$\int_0^3 (x^2 + x) dx = \frac{28}{2} = 14$$

Evaluate the integral

$$\int_0^3 (x+1)dx$$

Using Simpson's 3/8 rule

Answer ([Please click here to Add Answer](#))

X	0	1	2	3
Y=(x+1)	1	2	3	4

$$f(x, y) = yx^2, y(0) = 1 \text{ and } h = 0.1, x_0 = 0$$

$$k_1 = hf(x_0, y_0) = 0.1(1(0^2)) = 0$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= 0.1\left(0 + \frac{0.1}{2}, 1 + \frac{0}{2}\right)$$

$$= 0.1(0.05, 1) = 0.1(0.05 + 1) = 0.105$$

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2)$$

$$y_1 = 1 + \frac{1}{6}(0 + 2(0.105)) = 1 + \frac{0.21}{6} = 1.035$$

Find the dominant eigenvalue and the corresponding eigenvector of the matrix

$$\begin{bmatrix} 8 & 1 & 2 \\ 0 & 10 & -1 \\ 6 & 2 & 15 \end{bmatrix}$$

by Power Method with vector $V^{(0)} = (1, 1, 1)^T$ as the initial vector.

Answer ([Please click here to Add Answer](#))

vector $v^0 = (1, 1, 1)^T$ as the initial vector. Then, compute first iteration

$$u^1 = [A]v^0 = \begin{bmatrix} 8 & 1 & 2 \\ 0 & 10 & -1 \\ 6 & 2 & 15 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 8+1+2 \\ 0+10-1 \\ 6+2+15 \end{pmatrix} = \begin{pmatrix} 11 \\ 9 \\ 23 \end{pmatrix}$$

$$u^1 = 23 \begin{pmatrix} \frac{11}{23} \\ \frac{9}{23} \\ 1 \end{pmatrix} = q_1 v^1$$

second iteration

$$u^2 = [A]v^1 = \begin{bmatrix} 8 & 1 & 2 \\ 0 & 10 & -1 \\ 6 & 2 & 15 \end{bmatrix} \begin{pmatrix} \frac{11}{23} \\ \frac{9}{23} \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{88}{23} + \frac{9}{23} + 2 \\ 0 + \frac{90}{23} - 1 \\ \frac{66}{23} + \frac{18}{23} + 15 \end{pmatrix} = \begin{pmatrix} \frac{143}{23} \\ \frac{67}{23} \\ \frac{429}{23} \end{pmatrix}$$

$$u^2 = 23 \begin{pmatrix} \frac{143}{23} \\ \frac{67}{23} \\ \frac{429}{23} \end{pmatrix} = q_2 v^2$$

third iteration

$$u^3 = [A]v^2 = \begin{bmatrix} 8 & 1 & 2 \\ 0 & 10 & -1 \\ 6 & 2 & 15 \end{bmatrix} \begin{pmatrix} \frac{143}{23} \\ \frac{67}{23} \\ \frac{429}{23} \end{pmatrix} = \begin{pmatrix} 8+1+2 \\ 0+10-1 \\ 6+2+15 \end{pmatrix} \begin{pmatrix} \frac{143}{23} \\ \frac{67}{23} \\ \frac{429}{23} \end{pmatrix} = \begin{pmatrix} \frac{1573}{9867} \\ \frac{603}{9867} \\ 1.0 \end{pmatrix}$$

$$u^3 = 9867 \begin{pmatrix} \frac{1573}{9867} \\ \frac{603}{9867} \\ 1.0 \end{pmatrix} = 9867 \begin{pmatrix} 0.0159 \\ 0.0611 \\ 1.0 \end{pmatrix} = q_3 v^3$$

The largest eigen-value and the corresponding eigenvector as accurate to two decimals are

$$\lambda = 9867 \text{ and } (X) = \begin{pmatrix} 0.01 \\ 0.06 \\ 1.0 \end{pmatrix}$$

Question No : 51 of 52

From the following table of values, construct backward difference table.

x	0.1	0.2	0.3	0.4
$f(x)$	1.10517	1.22140	1.34986	1.49182

Answer ([Please click here to Add Answer](#))

X	F(x)	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$
0.1	1.10517			
0.2	1.22140	0.11623		
0.3	1.34986	0.12846	0.01223	
0.4	1.49182	0.14196	0.0135	0.00127

Evaluate the integral

$$\int_0^4 (x^2 + 1) dx$$

Using Simpson's 1/3 rule

Take $h=1$

Answer ([Please click here to Add Answer](#))

X	0	1	2	3	4
$Y=(x^2 + 1)$	1	2	5	10	17

$$\int_0^4 (x^2 + 1) dx = \frac{h}{3} (y_0 + y_4 + 4(y_1 + y_3) + 2(y_2))$$

$$\int_0^4 (x^2 + 1) dx = \frac{1}{3} (1 + 17 + 4(2 + 10) + 2(5))$$

$$\int_0^4 (x^2 + 1) dx = \frac{1}{3} (18 + 48 + 10)$$

$$\int_0^4 (x^2 + 1) dx = \frac{1}{3} (76) = \frac{76}{3} = 25.33$$